

“Evaluation of Two Approaches for Rainfall Prediction in the Junín Region: A Balanced Binary Classification Model and a Neural Network Classifier Model”

Authors: Lino, P., Calderón, C., Chavarria, M., Izarra, L., Mascco, P.

Abstract- Extreme rainfall in the Junín region is a critical climatic factor for the agricultural sector, particularly in relation to irrigation planning. This study addresses the challenge of classifying hourly precipitation into a severely imbalanced binary classification problem (No/Light Rain vs. Moderate/Heavy Rain), utilizing meteorological and irradiance data from the IGP (2021–2023) filtered for the peak rainy months. We evaluated the performance of a Balanced Statistical Model (Logistic Regression) compared to a Neural Network Classifier (MLPClassifier). To mitigate class imbalance, the SMOTE (Synthetic Minority Over-sampling Technique) method was applied to the training set, prioritizing the maximization of Recall for the critical minority class. The MLPClassifier demonstrated superior performance, achieving a Balanced Accuracy of 0.87 and a Recall of 0.94 for the moderate/heavy rain class. This result suggests that advanced non-linear models combined with oversampling techniques are highly effective for identifying rare and critical precipitation events, validating their use in early warning systems that prioritize risk detection over minimizing false alarms.

1. Introduction

Global climate variability has intensified the frequency and severity of extreme weather events. Consequently, the accurate prediction of precipitation has become an urgent priority for effective disaster management and water resource planning worldwide. The Andean region of Peru, and specifically the department of Junín, is particularly vulnerable to these climatic shifts. Within this context, extreme rainfall is a critical climatic factor for the agricultural sector, directly influencing irrigation planning and increasing the risk of crop diseases. The wet season, concentrated between the months of December, January, February, and March, is a period of heightened hydrological criticality for the Central Andes.

Traditionally, meteorological forecasting has relied on numerical models, which, while robust, often face limitations in temporal resolution (hourly data) and representing the complex topography of the Andean mountain range. Recently, Machine Learning (ML) models have emerged as a promising alternative, demonstrating success in capturing non-linear patterns within climatic time series. However, applying ML to the prediction of extreme rainfall presents an inherent challenge: severe class imbalance. Events of moderate or heavy precipitation are statistically rare compared to days with light or no rain, a factor that can bias

models toward the majority class and compromise performance in detecting critical events. A gap remains in the literature regarding a direct and balanced comparative evaluation between advanced linear and non-linear models tailored to address this specific imbalance problem in the Junín region.

Therefore, the primary objective of this study is to evaluate and compare two modeling approaches—a balanced binary classification model (Logistic Regression) and a Neural Network Classifier (MLPClassifier)—for predicting moderate to heavy precipitation events in the Junín Region, utilizing hourly IGP data from 2021 to 2023. The study makes a key methodological contribution by applying the SMOTE technique to the training set to mitigate class imbalance and focusing the evaluation on Recall (sensitivity) for the extreme precipitation class. The findings aim to determine the most suitable technique for implementing reliable early warning systems in the agricultural sector, providing an essential tool for local water resource risk management.

2. Related Work

Below is a summary of scientific articles related to the objective of this study.



Title: Monthly Precipitation Prediction Using Artificial Neural Networks for the Cali River Basin, Colombia

The authors examined how Artificial Neural Networks (ANNs) could be used to forecast monthly rainfall in the Cali River basin. They worked with data from 35 meteorological stations covering 1972–2016. Their Multilayer Perceptron (MLP) model used a hyperbolic activation function and was trained for nearly 6 000 iterations, achieving close to 98 percent accuracy. The results confirmed that neural models can represent complex, non-linear rainfall patterns that arise from seasonal behavior.

Even so, it seems that the model struggles to capture short-term variations and has difficulty dealing with imbalanced datasets, a common issue in regions like Junín, where about 95 percent of records show “no rain.” Working at a monthly scale also limits the model’s applicability for operational purposes. In this regard, the present study extends their idea by adopting an MLP-based neural architecture to classify hourly rainfall events and by introducing imbalance-handling techniques that make the approach more robust for sparse meteorological data.

Title: A Methodology Based on Random Forest to Estimate Precipitation Return Periods: A Comparative Analysis with Probability Density Functions in Arequipa, Peru.

In their research, the authors evaluated the Random Forest (RF) algorithm against classical Probability Density Functions (PDFs) such as Gumbel and Generalized Extreme Value (GEV) for estimating precipitation return periods in Arequipa, Peru. They used observations from 26 weather stations covering almost five decades (1965–2013). RF achieved higher accuracy in roughly two-thirds of the cases, showing stability against noise and strong generalization over return periods from 2 to 100 years.

The authors demonstrated that machine-learning methods can outperform traditional statistical approaches for extreme rainfall estimation. However, their goal was mainly the analysis of frequency rather than prediction of occurrence, and the issue of imbalanced data was not addressed. Our interpretation is that their results still support the potential of learning-based approaches for Peruvian contexts. The present study therefore moves toward ANN-based classification using an MLP network to forecast hourly rainfall events while explicitly examining the effect of data imbalance on predictive performance.

Title: Climate Variability and Its Impact on Food Security Crop Yields in the Highlands of Junín

The author investigated how climate variability—mainly temperature, rainfall, and humidity—affects the yield of 27 key crops in the highlands of Junín between 2015 and 2022. Using binary logistic regression, Pearson correlations, and t-tests, the study quantified the relationship between climate fluctuations and agricultural production.

While the author did not apply predictive models, the findings provide empirical support for the current research. It is clear that variations in rainfall affect both productivity and food-security outcomes. In practice, this confirms the need for more granular rainfall prediction methods that can support decision-making in agriculture. The present work draws on this evidence by linking MLP-based rainfall classification with agricultural applications to improve resilience and planning in Andean regions.

Title: Advancing preeclampsia prediction: a tailored machine learning pipeline integrating resampling and ensemble models for handling imbalanced medical data

The authors proposed a machine-learning pipeline for early preeclampsia prediction under extreme class imbalance. They tested eight resampling techniques (SMOTE, ADASYN, and IWGMM among others) combined with six ensemble algorithms including Random Forest and Gradient Boosted Trees, generating more than 4 600 experimental configurations. The evaluation used G-mean, MCC, AP, and AUC metrics and was validated on three independent datasets.

Although the context was biomedical, the authors offered a methodological lesson that extends to other fields. They showed that the sequence between resampling and training is crucial for model stability when classes are unbalanced. This principle directly inspired our Balanced Binary Classification Model, which applies SMOTE only to the training subset to avoid data leakage. By translating this logic to climatology, the present study enhances the reliability of rainfall classification under imbalanced conditions.

Title: SMOTE for Handling imbalanced Data Problem: A Review

The authors reviewed the Synthetic Minority Oversampling Technique (SMOTE), explaining how it creates synthetic samples for minority classes by interpolating between each instance and its nearest neighbors. Compared to random oversampling, SMOTE reduces overfitting and refines the decision boundary between classes.



In this study, their review serves as both conceptual and methodological reference. It supports the use of SMOTE to correct the dominance of “no-rain” records—around 95 percent of our dataset—and justifies its integration into MLP-based classification models. Moreover, by applying a 2.5 mm threshold to define “moderate/heavy rain,” the current research introduces a practical criterion that captures the specific climatic characteristics of the Andes.

3. Methodology

■ Proposed Approach(es):

A. Datasets Description.

The information used to train the models was obtained from datasets available on the National Open Data Platform of the Government of Peru. These datasets originate from the Automatic Meteorological Station (EMA) and the BSRN Station for Solar-Earth Energy Balance. In total, these two datasets served as the foundation for the model training process and contain the following attributes:

VARIABLE	DESCRIPTION	TYPE
FECHA_CORTE	Data cut off date	Numeric
UBIGEO	Geographical Location Code denoted as 'DDppdd' (Department, Province, District)	Alphanumeric
YY	Year, año de registro	Numeric
MM	Month, mes de registro	Numeric
DY	Day, día de registro	Numeric
HH	Hour, hora promedio del registro	Numeric
TT	Air Temperature 2 meters (temperatura del aire a 2 metros) - [°C]	Numeric
HR	Relative Humidity (Humedad Relativa) - [%]	Numeric
RR	Precipitation (precipitación) - [mm]	Numeric
PP	Atmospheric pressure (presión atmosférica) - [hPa]	Numeric
FF	Wind speed (velocidad del aire) - [m/s]	Numeric
DD	Wind direction (dirección del aire) - [degrees]	Numeric

Table 1: Data Dictionary of the EMA Dataset

VARIABLE	DESCRIPTION	TYPE
FECHA_CORTE	Data cut off date	Numeric
UBIGEO	Geographical Location Code denoted as 'DDppdd' (Department, Province, District)	Alphanumeric
year	Year, año de registro	Numeric
month	Month, mes de registro	Numeric
day	Day, día de registro	Numeric
hour	Hour, hora promedio del registro	Numeric
incide_LW_rad	Incident longwave irradiance (irradiancia incidente de onda larga) $W m^{-2}$	Numeric
emite_LW_rad	Emitted longwave irradiance (irradiancia emitida de onda larga) $W m^{-2}$	Numeric
direct_SW_rad	Direct shortwave irradiance (irradiancia directa de onda corta) $W m^{-2}$	Numeric
global_SW_rad	Global shortwave irradiance (irradiancia global de onda corta) $W m^{-2}$	Numeric
diffus_SW_rad	Diffuse shortwave irradiance (irradiancia difusa de onda corta) $W m^{-2}$	Numeric
reflec_SW_rad	Reflected shortwave irradiance (irradiancia reflejada de onda corta) $W m^{-2}$	Numeric

Table 2: Data Dictionary of the BSRN Dataset

B. Data Preprocessing.

To begin the preprocessing, the two datasets were integrated. First, the year, month, day, and hour columns of the primary dataset (EMA) were renamed to align with those in the auxiliary dataset (BSRN). This allowed for a successful merge, resulting in a single, unified dataset for subsequent analysis.

Next, the resultant dataset was cleaned by removing all rows containing null values. This step was deemed appropriate because the proportion of missing data in any column did not exceed 25%.

After this removal, a one-hour forward shift was applied to the precipitation variable RR, ensuring that each recorded value corresponds to the precipitation observed one hour after the respective timestamp.

A temporal filtering process was applied to focus the analysis on periods of peak rainfall activity. An exploratory analysis of the merged dataset revealed a marked seasonal pattern, with the vast majority of



precipitation concentrated in the months of December, January, February, and March. Therefore, to improve the model's ability to learn patterns associated with rainfall and reduce the high proportion of zero values in the target variable, the analysis was limited to this four month period, which represents the core of the rainy season. "The SAMS shows a seasonal cycle with an onset during October, a mature phase between December and February, and demise in April" (as cited in Giráldez et al., 2025, p. 4). We've decided to include March because of the quantity of precipitation in this month.

To address multicollinearity and reduce feature redundancy, a correlation matrix was computed on the initial set of variables from the merged dataset. Instead of applying a rigid numerical threshold, a hybrid criterion combining statistical correlation with domain knowledge was used. This procedure led to the exclusion of variables with clear physical redundancies, namely 'diffus_SW_rad', 'global_SW_rad', and 'reflec_SW_rad'. Conversely, variables that were strongly correlated but represent distinct physical processes, such as 'T' (Air Temperature) and 'emitte_LW_rad' (Emitted Long-Wave Radiation), were both retained. The resulting feature set exhibits low inter-feature correlation, ensuring a robust and high-quality dataset for training the models.

The primary target variable, Precipitación ('RR'), was transformed into a binary variable.

This approach transforms the variable into two categories: "No Rain/Light" (class 0, less than 2.5 mm), and "Moderate/Heavy" (class 1, greater than or equal to 2.5 mm). The 2.5 mm threshold was adapted from the rainfall intensity classification recommended by the World Meteorological Organization (WMO) in its *Guide to Meteorological Instruments and Methods of Observation*, which identifies 2.5 mm/h as the upper limit for light precipitation (WMO, 2024). By adapting this internationally recognized standard to an accumulation value, we ensure the discretization reflects a physically meaningful boundary between minor and more substantial rainfall events, which are of greater relevance to agricultural impact.

The binarization of the target variable resulted in a dataset with a significant class imbalance, where "No Rain/Light" instances formed a large majority. To address this, the dataset was first split into training (75%) and test (25%) sets. Following this split, the SMOTE (Synthetic Minority Over-sampling Technique) was employed exclusively on the training data. Unlike simple over-sampling, which duplicates existing records, SMOTE generates new synthetic samples for the minority category. The algorithm operates by selecting a minority class instance and creating a new synthetic data point along the line segment that connects it to one of its

nearest neighbors (Chawla et al., 2002). This procedure is crucial as it prevents synthetic information from leaking into the test set, thereby preserving its integrity as an unbiased benchmark for evaluating the model's final performance.

C. Modeling Approaches.

Balanced Binary Classification Model:

The development of this model followed a two-phase process. First, several algorithms were compared to determine which offered the best performance for the specific objectives of this study. Second, the selected model was trained and evaluated under balanced data conditions created with SMOTE.

The algorithms were trained on the SMOTE-balanced training data: Logistic Regression, 5-Nearest Neighbors, 10-Nearest Neighbors, CART (Decision Tree) with default depth, CART (max_depth=20), and XGBoost.

A 10-fold cross-validation process was used to compare their performance. While XGBoost obtained the highest overall Balanced Accuracy, its recall for class 1 was notably low (0.28). In contrast, Logistic Regression achieved a *Balanced Accuracy of 0.805* and a *Recall of 0.83* for the rainfall class, clearly outperforming the other models in detecting relevant precipitation events.

Given the practical importance of capturing all potential rain occurrences, Logistic Regression was chosen as the final classifier for this approach.

Once selected, the Logistic Regression model was implemented using *scikit-learn's* standard configuration. This choice was guided by the model's reliable convergence, interpretability, and consistent results under the SMOTE-resampled dataset.

Neural Network Classifier model:

After testing several traditional algorithms, a second model was built using a neural network to see if it could recognize rainfall patterns that linear methods might overlook. This experiment was carried out with the *Multilayer Perceptron (MLPClassifier)* from the *scikit-learn* library.

Neural networks were chosen mainly because weather variables—like humidity, temperature, wind, and solar radiation—tend to interact in complicated ways that are not easily captured by simple classifiers.

The network kept a practical feed-forward design with three hidden layers made up of 100, 30, and 15



neurons. Each layer used the ReLU activation function, which helps the model learn faster by avoiding saturation problems.

The output node relied on a sigmoid activation to produce probabilities between 0 and 1, marking the likelihood of rainfall.

Training was done with the Adam optimizer and limited to 1 000 iterations to prevent the model from running indefinitely.

A fixed random seed was set so the results could be reproduced later on the same data.

D. Evaluation Metrics

Assessing the performance of a rainfall prediction model requires more than just checking how often its predictions are correct. In this study, the data showed a strong imbalance, most samples represented *no rain*, while only a small portion recorded actual precipitation.

To prevent this, several complementary metrics were used: Balanced Accuracy, Precision, Recall, and F1-Score.

Balanced Accuracy (BA)

Balanced Accuracy measures how well the classifier performs across both classes by averaging the recall of each one. In this research, BA served as a key indicator of overall model balance, ensuring that predictions for both rain and no-rain events were evaluated on equal terms.

Precision

Precision represents how often the model's positive predictions were correct. Although a high precision is desirable, in early-warning systems it is less critical than recall.

Recall (Sensitivity)

Recall measures the model's ability to identify all true positive instances. For this project, maximizing the recall for the rain class (class 1) was the primary objective. This prioritization stems from the high cost associated with a False Negative in an agricultural context: a missed rainfall prediction can lead to significant crop damage, whereas a false alarm represents a minor operational inconvenience. Therefore, Recall was the decisive metric used to select the final model, directly aligning the evaluation with the practical need to minimize critical prediction errors.

F1-Score

The F1-Score combines precision and recall into a single number using their harmonic mean.

This balance helps evaluate how well a model detects rainfall without producing too many false

positives. In this work, the F1-Score offered an additional way to judge how each model managed the trade-off between sensitivity and reliability.

4. Experimentation and Results

■ Experimental Setup:

The models were developed in Python, using Google Colab as the development environment. Key libraries, including scikit-learn, were utilized for model implementation and evaluation.

The initial training set contained 4958 instances of class 0 and just 54 instances of class 1 while the test set contained 1653 instances of class 0 and 18 instances of class 1. SMOTE was applied to address the imbalance in the training set to create a balanced training set with 4958 instances of class 0 and class 1, too.

After completing the data preprocessing stage, each approach was implemented as follows:

■ Results:

A. Balanced Binary Classification Model:

Among the evaluated models, XGBoost achieved the highest *balanced accuracy* (0.9799 ± 0.0024), although the Logistic Regression (LogReg) model showed a lower average accuracy (0.8075 ± 0.0171) during cross-validation, it achieved the best performance during the training phase of the final model, outperforming the classifier based on XGBoost. While other algorithms were also evaluated, their results were not as central to the final conclusions.

Logistic Regression VS XGBoost

Balanced Accuracy Score in Models using SMOTE:

LogReg: 0.8050514216575922

XGBoost: 0.6189251865295422

	True Negative	False Positive	False Negative	True Positive
LogReg	1284	369	3	15
XGBoost	1587	66	13	5

Table 3: Confusion Matrix Comparative in Logistic Regression VS XGBoost .

	Precision	Recall	f1-score
LogReg (0)	1.00	0.78	0.87

XGBoost (0)	0.99	0.96	0.98
LogReg (1)	0.04	0.83	0.07
XGBoost (1)	0.07	0.28	0.11

Table 4: Classification Report Comparative in Logistic Regression VS XGBoost.

Logistic Regression (with SMOTE) recovers 15 of 18 heavy-rain cases (recall = 0.83), with BA = 0.805, precision = 0.04, and F1 = 0.07 for Class 1. The confusion matrix (TN = 1284, FP = 369, FN = 3, TP = 15) shows the cost of that sensitivity: many false positives that depress precision and F1. XGBoost identifies 5 of 18 events (recall = 0.28), with BA = 0.619, precision = 0.07, and F1 = 0.11 (TN = 1587, FP = 66, FN = 13, TP = 5).

The Logistic Regression algorithm was selected as the final model because it provided the best performance on the most critical metric for this problem: Recall for the "Moderate/Heavy Rain" class, achieving a score of 0.83. In this early-warning context, the cost of a false negative (missing an actual rainfall event) is significantly higher than the cost of a false positive (issuing a false alarm). Therefore, maximizing the model's ability to capture as many true rain events as possible was the top priority, making the recall score the decisive factor in the selection process.

Logistic Regression with SMOTE VS Logistic Regression without SMOTE

Balanced Accuracy Score in Logistic Regression:

LogReg with SMOTE: 0.8050514216575922

LogReg without SMOTE: 0.794464098003629

	True Negative	False Positive	False Negative	True Positive
LogReg with SMOTE	1284	369	3	15
LogReg without SMOTE	1249	404	3	15

Table 5: Confusion Matrix Comparative in Logistic Regression with SMOTE VS Logistic Regression without SMOTE.

	Precision	Recall	f1-score
LogReg with SMOTE (0)	1.00	0.78	0.87

LogReg without SMOTE (0)	1.00	0.76	0.86
LogReg with SMOTE (1)	0.04	0.83	0.07
LogReg without SMOTE (1)	0.04	0.83	0.07

Table 6: Classification Report Comparative in Logistic Regression with SMOTE VS Logistic Regression without SMOTE.

Both versions reach the same Class-1 recall (0.83, TP = 15, FN = 3). SMOTE does not raise sensitivity here; it mainly trims false positives (404 → 369), nudging balanced accuracy from 0.794 to 0.805 and slightly improving specificity. Class-1 precision remains 0.04 and F1 0.07 in both cases. In practice, Logistic Regression already finds most events; SMOTE helps by reducing spurious positives rather than by increasing true positives.

B. Neural Network Classifier model:

MLP with SMOTE VS MLP without SMOTE

Balanced Accuracy Score in MLP:

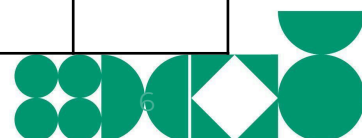
MLP with SMOTE: 0.8675640250050414

MLP without SMOTE: 0.5

	True Negative	False Positive	False Negative	True Positive
MLP with SMOTE	1307	346	1	17
MLP without SMOTE	1653	0	18	0

Table 7: Confusion Matrix Comparative in MLP with SMOTE VS MLP without SMOTE.

	Precision	Recall	f1-score
MLP with SMOTE (0)	1.00	0.79	0.88
MLP without SMOTE (0)	0.99	1.00	0.99
MLP with SMOTE (1)	0.05	0.94	0.09



MLP without SMOTE (1)	0.00	0.00	0.00
-----------------------------	------	------	------

Table 8: Classification Report Comparative in MLP with SMOTE vs MLP without SMOTE

With SMOTE, the MLP attains the highest Class-1 recall in the study (recall = 0.94), together with BA = 0.868, precision = 0.05, and F1 = 0.09 (TN = 1307, FP = 346, FN = 1, TP = 17). The precision and F1 remain modest because false positives persist, but the gain in sensitivity is decisive when missed detections are costly. Without SMOTE, the same network collapses to the majority class (TN = 1653, FP = 0, FN = 18, TP = 0), yielding recall = 0.00, precision = 0.00, F1 = 0.00, and BA = 0.500. Class balancing is therefore essential for the MLP in this dataset.

C. Performance Comparison of Rainfall Classifiers With and Without SMOTE Resampling:

	recall (1)	f1-score (1)	Balanced accuracy	Precision (1)
Logistic Regression with Smote	0.83	0.07	0.805	0.04
Logistic Regression without Smote	0.83	0.07	0.794	0.04
MLP with Smote	0.94	0.09	0.868	0.05
MLP without Smote	0.00	0.00	0.5	0.00

Table 9: Performance Comparison of Rainfall Classifiers With and Without SMOTE Resampling

This table presents a comparative evaluation of two machine learning models (Logistic Regression and Multi-Layer Perceptron) under two training conditions: with and without the application of the Synthetic Minority Over-sampling Technique (SMOTE). The models were assessed using four key performance metrics focused on the minority class (moderate/heavy rainfall): Recall, F1-Score, Balanced Accuracy, and Precision. The results highlight the impact of SMOTE on improving the models' ability to detect rare precipitation events, particularly in terms of recall and overall balance. Notably, the MLP model trained with SMOTE achieved the highest recall (0.94) and balanced accuracy (0.868),

confirming its suitability for early-warning systems where sensitivity to rainfall events is prioritized over precision.

5. Conclusion

In summary, this study compared two approaches for rainfall prediction: a binary classification model and a multi-layer perceptron (MLP) neural network, both trained with and without the SMOTE technique to address class imbalance.

SMOTE had an impact on both models, improving their ability to detect the minority class (rainfall events). For the binary classification model with LogisticRegression, using SMOTE increased the balanced accuracy from 0.79 to 0.81. For the MLP model, the effect of SMOTE was even more pronounced: without SMOTE, the MLP completely failed to predict rainfall (balanced accuracy of 0.50 and recall of 0.00 for the minority class), whereas with SMOTE it achieved a balanced accuracy of 0.87 and a recall of 0.94 for the minority class.

These results indicate that the MLP neural network with SMOTE is the most effective approach, capturing rare but critical rainfall events while maintaining good overall predictive balance. Therefore, SMOTE is essential for improving model performance in highly imbalanced datasets, and the MLP provides the optimal predictive capability among the tested models.

6. Suggestions for Future Work

Improvement of Data Preprocessing:

- Implement advanced imputation techniques to handle missing or erroneous values in meteorological variables.
- Incorporate satellite or radar-based meteorological data sources to enhance the quality and spatial resolution of the dataset.

Evaluation of Deep Learning Models:

- Explore architectures such as LSTM, GRU, or Transformers to model temporal sequences and capture complex precipitation patterns.
- Compare their performance against the balanced binary and neural network classifier approaches.

Validation in Other Agricultural Regions of Peru:

- Replicate the approach in other regions with distinct agroclimatic characteristics, such as Arequipa, Cusco, or Piura, to assess its generalizability.

Model Explainability and Trust:



- Apply tools such as SHAP or LIME to interpret model decisions, which is crucial for adoption by farmers and institutions.

Multi-Horizon Precipitation Forecasting:

- Develop models capable of predicting not only hourly precipitation but also short- and medium-term rainfall forecasts.

7. Ethical Implications

Responsible Use of Data:

- It is essential to ensure principles such as data quality and privacy in meteorological datasets. Although these are not personal data, poor processing can lead to misguided agricultural decisions. To address this issue, it is crucial to implement automatic data validation processes, real-time quality control, and auditing mechanisms that guarantee the integrity and reliability of the information used by predictive models.

Risk of Blind Trust in Automated Models:

- If farmers or institutions rely solely on predictions without understanding their limitations or error margins, there is a risk of making inappropriate decisions that could harm crops. To prevent this, training in model interpretation should be promoted, and uncertainty reports or error margins should be required alongside predictions.

Responsibility and Transparency in Predictions:

- Models must be explainable and auditable to avoid arbitrary or unjustified decisions in resource allocation. This requires the use of interpretability techniques such as SHAP or LIME, maintaining detailed records of training and validation processes, and documenting model decisions to facilitate review by experts and stakeholders.

Ethics of Bias and Representativeness:

- If the model is trained on limited or non-representative data from the region, it may introduce biases that disproportionately affect certain areas or crop types. Ethically, such biases must be identified and corrected. This involves conducting fairness analyses of model performance across different subregions or crop types and supplementing the data with more diverse sources.

Long-Term Sustainability of the System:

- If the system is implemented without a sustainability plan, it may fail over time, creating dependency on a technology that eventually becomes obsolete. Ethical

design requires consideration of long-term impact and local capacity to maintain the tool. This includes incorporating a maintenance plan from the outset, providing ongoing training for local users, using open and scalable technologies, and integrating with existing technical capabilities in the region.

8. Project Repository Link

<https://github.com/MrLinoP/Modelos-Predictivos-Precipitacion>

9. Author Contribution Statement

Lino, J.: Formulation of the hypothesis, dataset acquisition and translation of the original Spanish version of the document into English. Additionally, contributed to the drafting of the abstract, introduction, and experimental tables, and partial drafting of the Experimentation and Results section.

Calderón, C.: Drafting of the methodology, hypothesis, objective, and proposed approach. Synthesis of two related works. Development of the methodology and code (both approaches).

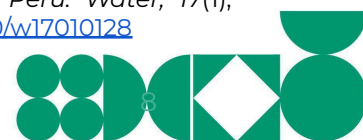
Chavarria, M.: Addition of two academic sources for the related work section. Contribution of methodological insights. Drafting of the experimentation and results, conclusions, future work suggestions, and ethical implications. Modifications in methodology and experimentation and results sections.

Izarra, L.: Contributed to the drafting of section Related Works. Contributed to the drafting of section Methodology. Drafting of the experimentation and results, conclusions, future work suggestions, and ethical implications. In addition, I modified the collab notebook.

Mascco, P.: Acquisition of academic sources used as guidance for the model, drafting of the introduction, and contribution of ideas to the code. Drafting of the experimentation and results, conclusions, future work suggestions, and ethical implications.

10. References

[1]. Anco-Valdivia, J., Valencia-Félix, S., Espinoza Vigil, A. J., Anco, G., Booker, J. D., Juárez-Quispe, J., & Rojas-Chura, E. (2025). *A methodology based on random forest to estimate precipitation return periods: A comparative analysis with probability density functions in Arequipa, Peru*. *Water*, 17(1), Article 128. <https://doi.org/10.3390/w17010128>



[2]. Arif, A. (2020). *How cross-validation works in machine learning*. Dataaspirant. <https://dataaspirant.com/cross-validation/>

[3]. Galarnyk, M. (s.f.). *Train test split: What it means and how to use it*. Built In. <https://builtin.com/data-science/train-test-split>

[4]. Instituto Nacional de Defensa Civil (INDECI). (2023). *Compendio estadístico del INDECI 2023: Gestión reactiva* (pp. 43–44). Lima, Perú: Autor. <https://cdn.www.gob.pe/uploads/document/file/5591372/4965310-compendio-final-af-2023-indeci.pdf>

[5]. Ma, Y., Lv, H., Ma, Y., Wang, X., Lv, L., Liang, X., & Wang, L. (2025). Advancing preeclampsia prediction: A tailored machine learning pipeline integrating resampling and ensemble models for handling imbalanced medical data. *BioData Mining*, 18(25), 1–16. <https://doi.org/10.1186/s13040-025-00440-1>

[6]. Montenegro-Murillo, D., Pérez-Ortiz, M., & Vargas-Franco, V. (2019). Predicción de precipitación mensual mediante redes neuronales artificiales para la cuenca del río Cali, Colombia. *Revista DYNA*, 86(211), 122–130. <https://revistas.unal.edu.co/index.php/dyna/article/view/76079/73107>

[7]. Paredes, V. (2023). *Variabilidad climática sobre el rendimiento de los cultivos de seguridad alimentaria en la sierra-Junín* (Tesis doctoral, Universidad Nacional del Centro del Perú, Facultad de Ciencias Forestales y del Ambiente). Repositorio Institucional UNCP. <https://repositorio.uncp.edu.pe/handle/20.500.12894/10334>

[8]. Pradipta, G., Wardoyo, R., Musdholifah, A., Sanjaya, I., & Ismail, M. (2021). SMOTE for handling imbalanced data problem: A review. In *2021 Sixth International Conference on Informatics and Computing (ICIC)* (pp. 1–8). IEEE. <https://doi.org/10.1109/ICIC54025.2021.9632912>

[9]. World Meteorological Organization (WMO). *Guide to Instruments and Methods of Observation* (WMO-No. 8), Volume I. Geneva, 2024. <https://library.wmo.int/idurl/4/68695>

[10] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*, 16, 321–357. <https://doi.org/10.1613/jair.953>

[11] Giráldez, L., Silva, Y., Zubieta, R., & Sulca, J. (2020). Change of the Rainfall Seasonality Over Central Peruvian Andes: Onset, End, Duration and Its Relationship With Large-Scale Atmospheric Circulation. *Climate*, 8(2), 23. <https://doi.org/10.3390/cli8020023>

